

BREAK INTO AEROSPACE

Rocket Lab

USA / New Zealand

Engineering Interview Preparation Guide

Targeted for the 2025–2026 Recruitment Cycle

Rocket Lab is one of the most distinctive companies in this entire series — a commercial launch company that is also building its own spacecraft, spacecraft components, and now hypersonic test vehicles, operating simultaneously in production (Electron) and clean-sheet development (Neutron/Archimedes) at a pace that is genuinely faster than any other company covered here. This guide consolidates the preparation tasks, signal areas, and strategic context needed to compete effectively across all relevant engineering roles.

PART I · WHAT ROCKET LAB ACTUALLY LOOKS FOR

By Seniority Level

- **Intern** — Rocket Lab’s intern hiring is predominantly driven by resume quality and a conversational technical interview. The intern interview process is largely non-technical in its first round — focused on resume walk-through and what was accomplished in previous internships — with a technical presentation required for interns joining engineering teams. The presentation format is specific: interns are asked to create a presentation on their most impressive technical accomplishment and present it to a panel of engineers with follow-up questions. The key filter at intern level is whether you have actually built or tested something that was not handed to you — whether your technical experience has real depth rather than surface participation. Rocket Lab looks for people who can adapt quickly, take ownership of a project from start to finish, and who exemplify the scrappy, mission-driven spirit the company values — still small enough that an individual can make meaningful contributions to big projects.

The ITAR citizenship requirement is a hard screen at every level. For US-based roles, US persons (citizens, permanent residents, refugees/asylees) are required. For New Zealand roles, the internship posting explicitly restricts eligibility to citizens of Australia, Japan, New Zealand, Switzerland, the European Union, and a small list of other US-friendly nations due to ITAR. This is not negotiable.

- **Mid-Level (~2–6 years)** — The typical Rocket Lab experienced hire process runs four rounds: a technical recruiter screen covering behavioral and resume, a hiring manager round, and technical interviews with engineers from the team. At mid-level, Rocket Lab is hiring engineers who can own a technical deliverable end-to-end — design it, analyse it, test it, and iterate on it — with minimal oversight and at high pace. Rocket Lab combines deep technical grilling with thorough behavioral evaluation and only about a third of candidates rate their interview experience as positive — a signal that the bar is high and the company is not just testing textbook knowledge but whether you really know your stuff and can apply it under pressure. The key cultural fit probe is whether you have operated autonomously under real constraints and tight timelines — whether you are genuinely hands-on or whether you describe work that was actually done by others.

The specific context of 2025–2026: Rocket Lab is simultaneously executing a high-cadence Electron launch programme (targeting 24 launches per year), developing the Neutron medium-lift reusable rocket (with the Archimedes engine completing first hot-fire testing at Stennis), scaling its Space Systems com-

ponent business, and executing on a \$515M US Space Force satellite contract awarded in 2024. The Archimedes engine uses an oxidiser-rich closed cycle operating at lower temperatures and pressures than gas generator engines to improve engine life and reusability. Engineers joining now are working across production programmes at operational cadence and advanced development simultaneously — a combination that is unusual even in new space.

- **Senior (5+ years, Lead, Principal)** — At senior level, Rocket Lab is looking for engineers who can independently own a major technical thread — a subsystem on Neutron, an avionics architecture decision for Photon, an engine design choice on Archimedes — and who can do this at Rocket Lab’s pace without needing external validation at every step. CEO Peter Beck has described Rocket Lab’s culture as: “We joke that a Rocket Lab year is like a dog year. One Rocket Lab year feels like five.” Senior engineers at Rocket Lab are expected to carry the velocity of the company — not manage it from a distance. The specific senior differentiator is whether you have personally closed a hard technical problem on a hardware system that either flew or was tested — whether your experience is from genuine ownership or from institutional support systems that insulated you from the consequence of being wrong.

Hands-On Signal Areas

Domain	Core Competencies Expected
Hardware Built and Tested	Rocket Lab is not just testing textbook knowledge — they want to see if you really know your stuff and can apply it under pressure, with a hands-on, get-it-done mindset. A candidate whose “design” project was run through analysis tools and handed to someone else to build is a weaker signal than one whose design was built, tested, revealed problems, was iterated on, and ultimately worked.
Electric Pump-Fed Propulsion	Electron’s Rutherford engine uses an electrically-driven turbopump — a design that eliminated the complex gas generator or preburner cycle, reducing parts count and improving reliability. Engineers who understand why this trade was made — and what it gives up versus what it enables — are demonstrating the first-principles reasoning Rocket Lab values.
Carbon Composite Structures	Electron’s body is a carbon composite monocoque structure. Rocket Lab is building carbon composite products for Neutron and sells composite products to external customers. Engineers with composite design, analysis, or manufacturing process experience are in persistent demand.
Embedded Software in C++	Technical interviews for software roles include coding challenges in C++ and embedded systems questions — and even hardware-focused roles frequently require familiarity with embedded software because Rocket Lab engineers often wear multiple hats.
Vertical Integration Mindset	Rocket Lab manufactures its own engines, avionics, carbon composite structures, launch vehicles, solar panels, reaction wheels, star trackers, separation systems, and spacecraft. Engineers who have worked within vertically integrated organisations are more immediately useful than those who know only their narrow specialisation.
US National Security Space	Rocket Lab became prime contractor for a \$515M United States Space Force satellite project in January 2024 — its largest contract to date. Engineers with US government satellite or launch vehicle experience are directly relevant to this growing part of Rocket Lab’s business.

How Rocket Lab’s Interview Format Differs from Standard Tech and Every Other Company in This Series

- **Four rounds, no assessment centre, no take-home project, no whiteboard algorithm exercises.** The process runs: technical recruiter screen (behavioral and resume) → hiring manager round → two or more technical rounds with engineers from the team. For interns, a technical presentation on their most

impressive accomplishment replaces some of the technical interview rounds. There is no group exercise, no structured competency scoring rubric, and no standardised assessment.

- **The technical interview is adversarial in a specific way.** Only about a third of candidates rate the Rocket Lab interview experience as positive — because the technical probing is deep, domain-specific, and not softened by social nicety. Interviewers are engineers working on the actual problems, and they will push until they find the boundary of your knowledge. The goal is not to make you feel good — it is to find out whether you actually know what you claim to know.
- **“Tie your answers back to Rocket Lab’s context” is the most important meta-preparation instruction.** Whatever your specialisation, tie your technical solutions back to Rocket Lab’s end goal — reliable launches and spacecraft. Interviewers respond to candidates who understand the mission and frame their technical expertise around enabling it. A thermal engineer who discusses heat management in the abstract is less compelling than one who references the specific thermal environments on Electron’s second stage or Neutron’s first-stage reentry.
- **The cultural fit probe is specific and probing.** Rocket Lab values agility and end-to-end ownership — engineers who wear multiple hats, adapt quickly, and take a project from concept through testing without needing someone else to define the path. “Tell me about a time you had to jump into a new role or learn something on the fly” is not a generic question at Rocket Lab — it is probing whether you have demonstrated the adaptability that Rocket Lab’s pace requires.
- **The Neutron context shapes every technical conversation in 2025–2026.** Neutron is Rocket Lab’s most ambitious development programme — a clean-sheet medium-lift reusable rocket with a novel Archimedes methane/LOX oxidiser-rich closed-cycle engine. Engineers who understand what Neutron is, why the Archimedes cycle was chosen over alternatives, and what the reusability architecture implies technically are demonstrating engagement with the company’s current most important technical challenge.
- **The US/NZ dual-geography creates a specific work context.** Rocket Lab’s launch vehicle manufacturing and launch operations are split between New Zealand (Electron assembly, engine testing, Launch Complex 1) and the US (Long Beach HQ/propulsion, Wallops LC-2, and other US facilities). Engineers at either location work across time zones with the other geography on a daily basis.

Honest caveat on format variation: Rocket Lab’s interview depth and format vary significantly by team and role. The Neutron propulsion team at Long Beach will probe very differently from the Space Systems avionics team or the New Zealand operations team. The four-round structure is typical but not universal. Filter Glassdoor reviews by specific role title wherever possible.

PART II · 20 SPECIFIC PREPARATION TASKS

Intern Level

1. Build a Technical Presentation on Your Most Impressive Hands-On Project — Structured for Adversarial Questioning by Engineers 4–5 h

Skill: The primary intern selection event at Rocket Lab

Objective: Interns are asked to create a presentation on their most impressive technical accomplishment and present it to a panel of engineers with follow-up questions. This is not a career summary — it is a deep technical dive into a single project. The follow-up questions will probe every non-obvious choice: why that material, why that algorithm, why that test configuration, what would fail first, what you got wrong and fixed. The presentation is the primary technical screen for interns.

4–5 h

Done Signal: You have a 15-minute presentation on your strongest technical project that covers: problem statement, your specific approach and why it was chosen over alternatives, the result, an anomaly or failure you encountered and how you resolved it, and what you would do differently. You have delivered it to at least two engineers outside your field who then asked three probing questions each, and you answered all six without retreating to vagueness.

1 h

2. Verify ITAR Eligibility for Your Target Rocket Lab Location (US or NZ)

Skill: Citizenship and export control self-assessment before applying

Objective: For New Zealand office roles, eligibility is restricted to citizens of Australia, Japan, New Zealand, Switzerland, the European Union, or a country that meets specific ITAR criteria due to US export control regulations. For US roles, US person status (citizen, permanent resident, refugee, or asylee) is required. Applying without meeting this requirement wastes everyone's time. If you are on a student visa or on OPT, you are very likely ineligible.

Done Signal: You have confirmed your citizenship status, verified which Rocket Lab sites and roles are accessible to you given your citizenship, and can answer the recruiter's ITAR eligibility question immediately and without hesitation.

2–3 h

3. Prepare for the Technical Recruiter Screen: Resume-Depth Probing on Every Project

Skill: Project narrative depth under conversational probing

Objective: The first technical round at Rocket Lab is a recruiter-led resume walk-through. Unlike many companies where the recruiter focuses only on cultural fit and timeline, Rocket Lab recruiters probe technical depth — they will ask what you actually did on a project versus what your team did, what design choices you made, and what the results were. Candidates who have padded their resume with peripheral contributions are caught quickly.

Done Signal: For every project on your resume, you can answer: what your specific individual contribution was (not the team's), what design choice you made and why, what the test result or outcome was, and what you would change if starting over. You can answer these in under 2 minutes per project without preparation time.

3 h

4. Study Electron's Technical Architecture — and Be Able to Explain It from First Principles

Skill: Rocket Lab product knowledge at the level of an engineering conversation

Objective: Electron is Rocket Lab's operational product — a small orbital rocket powered by nine Rutherford engines on the first stage and one vacuum-optimised Rutherford on the second stage. The Rutherford uses an electric pump-fed cycle: a novel departure from conventional gas generator or staged combustion designs. Interns who cannot describe why Rocket Lab chose an electric pump-fed cycle over a gas generator, or what advantage a carbon composite monocoque body offers, are demonstrating they have not engaged with the product they would be working on.

Done Signal: You can explain what an electric pump-fed cycle is and why Rocket Lab chose it over a gas generator (lower complexity, fewer parts, more predictable operation, easier scaling of thrust through pump speed), what LOX/RP-1 propellants offer versus methane or hydrogen, why the body is carbon composite rather than aluminium, and what the Rutherford engine's 3D-printed components enable for manufacturing pace. You can discuss all of this conversationally.

5. Prepare STAR-Method Behavioral Answers Specifically for Rocket Lab's Cultural Probes

3 h

Skill: Cultural fit demonstration for a scrappy, fast-moving, multi-hat environment

Objective: Rocket Lab places a strong emphasis on cultural fit and adaptability — interviewers probe how you work under challenging conditions, handle setbacks, and whether you exemplify the scrappy, mission-driven spirit. They want people who can adapt quickly and take ownership from start to finish. The specific questions probe: learning something new under time pressure, taking over a failing project, working without clear guidance, handling a significant technical failure, and doing more than your defined role required.

Done Signal: You have four STAR stories specifically for: a time you jumped into a new technical domain under time pressure and produced results, a time you took a project from concept through testing with minimal oversight, a time something failed and you diagnosed and fixed it, and a time you wore multiple hats to get a deliverable across the finish line. Each ends with a specific measured outcome.

6. Demonstrate Familiarity with Rocket Lab's Current Programme Context — Electron Cadence, HASTE, Neutron Development Status

4 h

Skill: Mission context that enables a peer conversation with the engineering team

Objective: Rocket Lab is not a generic aerospace employer — it is a specific company at a specific operational moment. The company is simultaneously executing Electron at 24-launches-per-year operational tempo, running HASTE hypersonic test missions, testing the Archimedes engine at Stennis, and scaling its Space Systems components business. A candidate who can say “I noticed Archimedes just completed its first hot-fire and I have questions about the ox-rich staged combustion design choice” is having a different conversation from one who only knows Electron exists.

Done Signal: You can describe Electron's current launch cadence and operational sites (Māhia and Wallops), explain what HASTE is and why hypersonic testing is a growing revenue stream, describe Neutron's architecture and the Archimedes engine's design choice, and name one specific engineering challenge in Neutron's development that you find technically interesting. You can do this without reading from notes.

Mid-Level (~2–6 years)

All intern tasks still apply. The presentation (task 1) must cover a professional project, not an academic one. The Electron and Neutron knowledge (tasks 4 and 6) must extend to specific engineering trade-offs, not just factual descriptions.

7. Prepare to Demonstrate First-Principles Reasoning on a Propulsion Engineering Problem Specific to Rocket Lab

5–6 h

Skill: Propulsion fundamentals applied to the Rutherford and Archimedes engine design context

Objective: Technical interviews probe whether you really know your stuff and can apply it under pressure — technical questions are framed in Rocket Lab's specific context. For propulsion roles: “Walk me through the trade-offs between an electric pump-fed cycle, a gas generator cycle, and a staged combustion cycle for a small orbital rocket” is a real question type. “Why did Rocket Lab choose LOX/RP-1 for Rutherford and methane/LOX for Archimedes?” is another. These questions have specific answers rooted in physics, manufacturability, and reuse considerations.

Done Signal: You can explain the electric pump-fed cycle, the gas generator cycle, and staged combustion from first principles — including specific impulse, complexity, failure modes, and reusability

implications of each. You can explain why LOX/RP-1 was chosen for Rutherford (density, handling, specific impulse at small scale) and why methane/LOX was chosen for Archimedes (reusability, cleanliness, specific impulse at medium scale). You can answer a follow-up question on each without going vague. 5–6 h

8. Study Neutron's Architecture at Engineering Depth — Reusable First Stage, Unibody Design, Archimedes Engine, and What "Clean Sheet" Means 4 h

Skill: Neutron development technical literacy — Rocket Lab's most consequential current programme

Objective: Neutron is the programme that a significant fraction of mid-level Rocket Lab hiring is building toward. Neutron features a reusable first stage, landing planned on a floating platform downrange, and nine Archimedes engines using an oxidiser-rich closed cycle operating at lower temperatures and pressures than gas generator engines to improve engine life and reusability. Engineers who understand what a clean-sheet medium-lift rocket development involves — the structural design choices, the propulsion cycle selection, the reuse architecture — are demonstrating they are ready to contribute to Neutron, not just observe it.

Done Signal: You can explain Neutron's key design features (40m tall, 4.5m fairing, 13–15 tonne LEO capacity, reusable first stage with ocean-platform recovery), describe why the Archimedes oxidiser-rich closed cycle was chosen for reusability over alternatives, explain what a unibody rocket structure means and what it trades against a more conventional section-based design, and describe one specific engineering challenge in Neutron's development that has not yet been publicly resolved.

9. Prepare a Story About End-to-End Ownership of a Hardware Project — from Design Requirement Through Test Result 2 h

Skill: The core ownership evidence Rocket Lab probes at mid-level

Objective: Agility and end-to-end ownership are the primary mid-level cultural filters at Rocket Lab — engineers who take a project from concept through testing without needing someone else to define the path are the profile they are hiring for. The story must cover the full arc: a requirement or problem, your design approach and the alternatives you rejected, the manufacturing or build process, the test or validation, the anomaly you encountered, and the final outcome. If you only designed something and handed it off, that is not end-to-end ownership in Rocket Lab's sense.

Done Signal: You have a story that covers the full arc — requirement, design (with a specific trade-off named), build or coding, test configuration, anomaly or unexpected result, fix, and verified outcome. Every step names your personal action, not the team's. The story demonstrates that you closed the loop — that the thing you designed was tested and either worked or was iterated until it did.

10. Study Carbon Composite Structures at a Working Engineering Level — Specifically for Launch Vehicle and Spacecraft Applications 4–5 h

Skill: Composite structures technical depth for Rocket Lab's signature manufacturing capability

Objective: Rocket Lab produces carbon composite structures for Electron, is building Neutron's composite structures at its Space Structures Complex in Middle River, and sells composite products to external customers. Carbon composite expertise is one of Rocket Lab's most distinctive manufacturing capabilities and a live hiring need. Engineers who understand composite lay-up design, cure cycle management, inspection methods (ultrasonic, thermal, CT), and failure modes (delamination, matrix cracking, impact damage) are directly relevant to multiple Rocket Lab production programmes.

4–5 h

Done Signal: You can explain what a carbon fibre reinforced polymer structure is, describe the difference between prepreg and wet lay-up manufacturing, explain what autoclave cure conditions control, describe the primary composite failure modes and how they are detected, and connect this to one specific Electron structural challenge (e.g., thermal cycling from cryogenic propellant fill to ascent aeroheating).

1–2 h

11. Prepare a Story About Diagnosing a Hardware Anomaly Under Time Pressure — Where the Root Cause Was Not Obvious

Skill: Test-anomaly investigation capability at operational pace

Objective: Rocket Lab is conducting Electron launches at a cadence of one every two to three weeks. When an anomaly appears during launch prep or in-flight, the engineering team must diagnose it rapidly, make a disposition decision, and resolve the root cause before the next mission. Engineers who have done this — who have diagnosed a real hardware failure with time pressure and incomplete information — are directly relevant to Rocket Lab's operational tempo.

Done Signal: You have a story where a hardware anomaly appeared under time pressure, the root cause was not immediately apparent, you designed and executed a diagnostic approach, the first hypothesis was wrong, and you resolved the root cause. The story names the specific failure symptom, the incorrect hypothesis, the diagnostic approach that corrected it, and the fix that was implemented.

2–3 h

12. Understand Rocket Lab's Space Systems Division — What Components They Manufacture and Why Vertical Integration Enables Their Launch Competitiveness

Skill: Space Systems business awareness for the fastest-growing part of Rocket Lab's revenue

Objective: Rocket Lab's Space Systems division manufactures star trackers, reaction wheels, solar panels, electric propulsion systems, software-defined radios, composite structures, separation systems, and EO/IR sensors — components that go into satellites built by third parties as well as Rocket Lab's own Photon spacecraft. This vertical integration means Rocket Lab can draw on in-house component expertise when designing Photon or Electron avionics, reducing schedule risk and enabling design iteration that external component vendors could not support.

Done Signal: You can name five Space Systems products and explain why Rocket Lab chose to manufacture them internally rather than buying from a supplier (for each, there is a control, schedule, or technology argument). You can explain how the Space Systems division's products connect to Rocket Lab's launch vehicle competitiveness — specifically how Photon spacecraft integration with Electron affects launch schedule and risk.

4–5 h

13. Prepare to Answer Domain-Specific Technical Questions That Are Explicitly Framed in Rocket Lab's Operational Context

Skill: Technical depth applied to Rocket Lab's specific hardware context

Objective: Interviewers expect you to tie your technical answers back to Rocket Lab's context — to show that you understand the end goal of reliable launches and spacecraft and frame your technical solutions around enabling that mission. Generic textbook answers — even correct ones — score lower than answers that connect the engineering principle to a specific Rocket Lab challenge. A thermal engineer answering “how do you manage thermal loads on a spacecraft component?” who references the specific thermal environment on Electron's second stage during coast phase and reentry is demonstrating mission-relevant engineering thinking.

Done Signal: For five technical questions in your specific engineering domain (propulsion, structures, thermal, avionics, software, etc.), you have prepared answers that explicitly reference a spe-

cific Rocket Lab vehicle, component, or operational scenario — not generic aerospace principles. You have spoken each answer aloud to confirm it sounds like an engineer discussing a specific problem, not a student reciting a definition. 4–5 h

Senior Level (5+ years, Lead, Principal)

All mid-level tasks still apply. The end-to-end ownership story (task 9) must involve a production system or programme that flew or was deployed — not a prototype or development artefact. The Neutron knowledge (task 8) must include a specific technical opinion on a design trade-off.

14. Prepare to Walk Through a Complete Propulsion or Vehicle Subsystem You Led — Including Anomalies, Design Changes, and Lessons Applied to the Next Design Iteration 4–6 h

Skill: System-level technical ownership at the pace and ambition Rocket Lab’s senior engineers require

Objective: A Rocket Lab year is like a dog year — the pace of iteration and development is 5x that of a conventional aerospace company. Senior engineers at Rocket Lab are expected to have closed technical loops quickly — to have owned a propulsion system, avionics architecture, structural design, or flight software module from requirements through flight, encountered failures, iterated on the design, and applied lessons forward. Candidates who describe long development cycles with many reviewers and approval steps are describing a pace that does not match Rocket Lab’s operating model.

Done Signal: For one propulsion subsystem, avionics architecture, or vehicle structural element you personally led, you can answer questions at every stage — the design requirements, the key trade-off made, the test programme, the first major anomaly, the design change driven by that anomaly, and how the lesson was applied to the next development cycle. You can describe the pace: how long from design to first test, and how many iterations before satisfactory performance was achieved.

15. Prepare a Technically Grounded Position on the Neutron Development Challenges Most Relevant to Your Engineering Domain 3–4 h

Skill: Forward engineering engagement at the level of Rocket Lab’s most active development

Objective: Neutron is the programme that will define Rocket Lab’s next decade. Senior engineers in every domain — propulsion, structures, avionics, GNC, software, manufacturing — have specific contributions to make to Neutron that are distinct from what they contributed to Electron. The senior candidate who can say “my experience with ox-rich staged combustion on [prior programme] is directly applicable to the Archimedes life prediction challenge during reuse” is having a peer-level technical conversation that demonstrates exactly the capability Rocket Lab needs.

Done Signal: You have a prepared 3-minute technical discussion of one specific engineering challenge in Neutron development — named specifically and connected to your engineering background. The challenge is technically grounded (not just “reusability is hard”), your position on the approach includes a view on what has to go right and what the primary failure mode risk is, and you can connect it to something you have personally worked on.

16. Prepare Examples of Making Significant Engineering Decisions with Incomplete Data Under Time Pressure — and Being Proved Right or Wrong 2–3 h

Skill: Decision quality under Rocket Lab’s operational pace

2–3 h

Objective: At Rocket Lab’s cadence, perfect data is rarely available when decisions must be made. The launch team decides whether to launch on schedule or hold based on partial sensor data, incomplete weather models, and engineering judgement. Senior engineers are expected to make these calls — and to have a track record of making them well. The ability to articulate the decision framework used (“I had these data points, made these assumptions, accepted this residual risk, and set this abort criterion”) is a senior-level capability.

Done Signal: You have two stories where you made a significant engineering decision with incomplete data under real time pressure — one where the outcome confirmed the decision was right, and one where the outcome revealed the decision was wrong. For each, you can articulate the data you had, the assumption that bridged the gap, the risk you accepted, and what you learned about your decision-making process.

3 h

17. Prepare to Discuss the First-Stage Reuse Architecture of Neutron — Specifically the Engineering Trade-Offs Compared to Falcon 9’s Approach

Skill: Reusable launch vehicle engineering at the level of a Rocket Lab senior engineer conversation

Objective: Neutron’s reusability architecture includes a first stage landing on a floating platform downrange, with the fairings and Archimedes engines all reusable. This is a specific design philosophy with specific engineering consequences: the ocean platform approach changes recovery logistics, re-entry thermal protection requirements, and propellant mass allocation for landing burns compared to landing on a fixed pad. Senior engineers who can engage with why Rocket Lab made these choices — and what the implications are for vehicle life, inspection, and refurbishment — are demonstrating reusability engineering depth.

Done Signal: You can explain the key differences between Rocket Lab’s Neutron recovery approach and Falcon 9’s (ocean platform vs. onshore landing, unibody rocket architecture vs. cylindrical stage, Archimedes ox-rich cycle vs. Merlin gas generator), describe what the floating platform approach means for first-stage design requirements (landing load cases, salt water exposure, crane lifting interface), and articulate one specific engineering challenge in the approach that has not yet been publicly resolved.

4 h

18. Build Quantitative Estimation Fluency for Launch Vehicle Performance Parameters — Specific to Small and Medium-Lift Class Vehicles

Skill: Rocket performance reasoning from first principles without a simulation tool

Objective: Senior Rocket Lab engineers sanity-check trajectory analyses, estimate payload capacity impact from design changes, and respond to “does this number make sense?” in fast-moving design reviews. For a launch vehicle company, this means: estimating delta-v budget from orbital mechanics, estimating payload fraction from Tsiolkovsky and a staging assumption, estimating Isp impact from a propellant change, or estimating landing burn propellant mass from a given deceleration requirement. None of these requires a trajectory simulation to estimate at order-of-magnitude.

Done Signal: You have worked through five launch vehicle performance estimation problems — spoken aloud, assumptions stated before numbers — arriving at order-of-magnitude answers with stated uncertainty. Examples: estimate the payload fraction of a two-stage LOX/RP-1 vehicle to LEO given typical Isp and mass fraction assumptions; estimate the propellant required to decelerate a 10,000 kg stage from entry velocity to zero at sea level. You can identify which assumption is most load-bearing in each.

All Levels

19. Prepare a Specific, Technically Grounded Answer to “Why Rocket Lab — and Why Now?”

1–2 h

Skill: Mission-specific motivation that engages with Rocket Lab’s specific moment in 2025–2026

Objective: Rocket Lab is looking for people who align with their mission of opening access to space and who have a hands-on, get-it-done mindset — genuine passion for the work is as important as technical mastery. The “why Rocket Lab” question is not an HR formality — it is a genuine probe of whether you have thought about what it means to work at a company simultaneously running operational launches and developing the next generation of rockets at unusually high pace. “Because I want to work in space” is not the answer. The expected answer names a specific Rocket Lab programme or technical challenge, explains why that specific challenge at this specific company is compelling, and demonstrates that you understand the operational context you are joining.

Done Signal: You have a 90-second answer that names a specific Rocket Lab programme (Neutron development, HASTE hypersonic, Archimedes engine development, Photon spacecraft for NASA), explains one technically specific aspect of it that you find genuinely interesting, connects your engineering background to contributing to it, and acknowledges the pace and culture you are signing up for. The answer could not be given unchanged to any other commercial launch company.

20. Prepare a Story About Moving Faster Than Your Organisation’s Process Allowed — and How You Managed the Balance Between Speed and Rigour

1–2 h

Skill: Pace calibration for Rocket Lab’s “dog year” operating tempo

Objective: Rocket Lab’s culture values speed. But Rocket Lab also launches rockets — and a launch vehicle failure is not an iteration event. CEO Peter Beck has described himself as both the “go-getter entrepreneur” and the “engineer realist” who wrestle with each other — resulting in a culture that is ambitious but pragmatic, and never bets the company on anything. Senior engineers at Rocket Lab are expected to have navigated this tension personally — to have examples of moving faster than conventional process would allow while maintaining the engineering rigour that safety-critical systems require.

Done Signal: You have a story about moving faster than your organisation’s standard process — you can name what was skipped or compressed, what residual risk you accepted and why, what monitoring or abort criteria you put in place to manage that risk, and what the outcome was. The story demonstrates both the velocity you brought and the engineering judgement that contained the risk — not just speed.

SUMMARY REFERENCE TABLE

#	Task Description	Target Level	Effort
1	Technical presentation on strongest hands-on project	All	4–5 h
2	ITAR/citizenship eligibility — US or NZ site confirmation	All	1 h
3	Resume-depth probing preparation — every project	All	2–3 h
4	Electron architecture — electric pump-fed, composites, Rutherford	All	3 h
5	STAR behavioral stories — scrappy, multi-hat, ownership	All	3 h
6	Programme context — Electron cadence, HASTE, Neutron status	All	4 h
7	Propulsion cycle first principles — Rutherford vs. Archimedes	Mid+	5–6 h
8	Neutron architecture — reusable first stage, Archimedes, unibody	Mid+	4 h
9	End-to-end hardware ownership story — requirement through test	Mid+	2 h
10	Carbon composite structures — lay-up, cure, failure modes	Mid+	4–5 h
11	Hardware anomaly diagnosis under time pressure	Mid+	1–2 h
12	Space Systems division vertical integration logic	Mid+	2–3 h
13	Domain technical questions framed in Rocket Lab context	Mid+	4–5 h
14	Subsystem ownership walk-through — anomaly-iterate-apply	Senior	4–6 h
15	Neutron development challenge — domain-specific technical position	Senior	3–4 h
16	Decision-under-incomplete-data — two stories, one right, one wrong	Senior	2–3 h
17	Neutron first-stage reuse architecture vs. Falcon 9 trade-offs	Senior	3 h
18	Launch vehicle performance estimation — five problems	Senior	4 h
19	“Why Rocket Lab — and why now?” — programme-specific answer	All	1–2 h
20	Moving faster than process allowed — speed vs. rigour balance	All	1–2 h

THE STRUCTURAL REALITY

Every other organisation in this guide has an established institutional framework — a certification process, a programme management structure, a government customer relationship, or an academic governance model — that creates a predictable pace of engineering work. Rocket Lab has deliberately built itself to operate without most of these frameworks. It manufactures its own rocket engines, structures, avionics, and spacecraft. It owns its own launch sites. It runs its own launch operations. It is developing a new rocket from scratch while simultaneously operating an existing one at increasing cadence.

The engineers who succeed at Rocket Lab are those who find the absence of institutional scaffolding energising rather than unnerving — who arrive with enough depth to make good decisions without requiring a committee to validate them, enough breadth to understand the systems their decisions affect, and enough genuine enthusiasm for the mission that the pace of a dog year does not exhaust them within the first six months.

The ones who struggle are those who confuse Rocket Lab's speed for a lack of rigour, or who confuse its informality for a lack of standards. The company launches rockets into orbit. When it fails, the failure is immediate, public, and costly. The scrappy culture and the engineering rigour are not opposites at Rocket Lab — they are the same thing.

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